

Research

Phenological responses to climate warming in temperate moths and butterflies: species traits predict future changes in voltinism

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Changes in the number of generations per year (voltinism) have been among the most common phenological responses to climate warming in insects inhabiting seasonal environments. Nevertheless, numerous species have maintained univoltine (one generation per year) phenology with increasing temperatures, indicating the involvement of phylogenetic, ecological or some other constraints on phenological change. I examined geographic variation in voltinism in moths and butterflies of northern Europe to identify species traits that might predispose species to univoltine/multivoltine phenology. I focused on species with a wide latitudinal distribution range (15 degrees as a minimum) which makes it unlikely that constraints imposed by season length could preclude multivoltinism across their distribution. Almost half of the 731 moth and butterfly species considered appear to have a single generation throughout their entire European range. A univoltine life-cycle across a wide latitudinal gradient suggests the presence of some constraint that makes additional generations either impossible or at least strongly disadvantageous, which will unlikely change with future climate warming. The scattered distribution of univoltine and multivoltine species across the lepidopteran phylogeny indicates that phylogenetic constraints do not strongly limit changes in voltinism, and the trait is open to ecologically-driven adaptive evolution. My data show that species with one generation per year are generally larger than multivoltine species, but size forms no absolute constraint to having multiple generations per year. Obligately univoltine species dominate among egg and adult overwinterers (life-histories typical of so-called spring-feeders), whereas species with capacity for multiple generations prevail among pupal overwinterers. Multivoltinism is also infrequent among species feeding on grasses, particularly in endophagous grass-feeders. Larval diet breadth has no discernible effect on voltinism. Given the diverse ecological consequences of voltinism and its changes, accounting for the species' capacity for multivoltinism may be a key to address future challenges in biodiversity conservation and pest management.

Keywords: body size, diapause, functional trait, latitudinal variation, phenology, voltinism

Introduction

The effects of ongoing climate warming range from shifts in species distribution (Parmesan et al. 1999) to changes in body size of the organisms (Gardner et al. 2011), with diverse consequences for populations, species, communities and ecosystems (Carter et al. 2018). Changes in species' phenologies are among the best documented impacts of climate warming (Parmesan 2006, 2007, Cohen et al. 2018, Kharouba et al. 2018, Duchenne et al. 2020). In short-lived organisms, such as insects and many other ectotherms, seasonally earlier start of development and activity, temporal mismatches in consumer–resource interactions, and increased number of generations per year have been the most reported phenological responses (Altermatt 2010a, Diamond et al. 2011, Pöyry et al. 2011, Roy et al. 2015, Kharouba et al. 2018, Visser and Gienapp 2019). Nevertheless, phenological sensitivity to climate change strongly varies among species (Primack et al. 2009, Altermatt 2010b, Valtonen et al. 2011, Kharouba et al. 2014, Végvári et al. 2015, Fric et al. 2020). As phenology affects most aspects of species' ecology (Forrest and Miller-Rushing 2010), identifying traits that associate with phenological sensitivity is necessary for making predictions about species responses to future climate changes.

In temperate environments, the length of the period with favourable ambient temperatures sets the limits for ectotherm growth and development, including the number of generations a species can potentially complete within a season. Consistently, the number of generations per year (= voltinism) typically decreases towards higher latitudes in species capable of multiple generations (Danilevsky et al. 1970, Roff 1980, Tauber et al. 1986, Zeuss et al. 2017). In insects, northern populations of many species have shifted from univoltine (i.e. one generation per year) to multivoltine (two or more generations per year) phenologies as a response to recent climate warming (Altermatt 2010a, Pöyry et al. 2011). Altermatt (2010b) has shown that shifts in voltinism in response to past temperature increases have occurred in a somewhat predictable manner, with certain traits predisposing species to an increase in voltinism. However, for the overwhelming majority of such species, the increase in voltinism is not unexpected as they have already been regularly multivoltine at lower latitudes of their distribution ranges. The transition area from a univoltine to a multivoltine life cycle in these species has just shifted towards higher latitudes with increasing temperatures. In such species, winter diapause is typically facultative, evolved to survive periods unfavourable for growth, development and reproduction (Danilevsky et al. 1970).

Nevertheless, numerous insect species, even those widely distributed across latitudes, have maintained a univoltine phenology throughout their ranges. As time constraints on development, imposed by seasonality, should be much more relaxed at lower latitudes of species' distribution ranges, season length is hardly a decisive factor in limiting voltinism in such species (Tammaru et al. 2001). This leads to the question of what species traits preclude producing multiple

generations per season. As species that are able to increase the number of seasonal generations have been suggested to become also more abundant with ongoing climate warming (Tobin et al. 2014, Macgregor et al. 2019), identifying 'truly' univoltine and potentially multivoltine species is also important from the perspectives of species conservation and pest control.

In this study, I analyse voltinism in European moths and butterflies to identify traits that associate with producing multiple generations. I use space-for-time substitution approach that is a widely used practice to infer future changes from contemporary spatial patterns (Hodgson et al. 2011, Blois et al. 2013). In particular, as the magnitude of ongoing climate warming has not necessarily been sufficient to identify obligately univoltine species, I rely on examining geographic differences in voltinism instead of tracking long-term changes in voltinism in response to recent climate warming. For the purpose of this study, I collected data on a subset of North European Macrolepidoptera (altogether 731 species) with a wide latitudinal distribution range (15° latitudinal degrees as a minimum). Such species experience a wide range of season lengths which makes it unlikely that constraints imposed by season length alone could preclude multivoltinism throughout entire distribution ranges. Consistently, I interpret the occurrence of multivoltinism at any part of a species range as an indication of the potential for species to increase the number of generations with climate warming. By contrast, univoltinism throughout the distribution range will indicate the presence of some constraint – phylogenetic, ecological, genetic or physiological – precluding such response. My primary focus in this study is on revealing phylogenetic and ecological constraints to voltinism.

My analysis of moths and butterflies was facilitated by their remarkable diversity in body size and ecological traits, such as host use patterns (diet breadth, host type) and the timing of life cycle. Practical considerations for focusing on Macrolepidoptera were also important: for most moths and butterflies, at least in Europe, ecological data are readily available. Thanks to regular monitoring schemes and public interest, the variable to be explained, i.e. the number of generations per season is well documented for most species throughout their range.

Material and methods

Species selection

To examine the determinants of species' capacity for multivoltinism, I compiled a list of Macrolepidoptera (Lasiocampoidea, Bombycoidea, Noctuoidea, Drepanoidea, Geometroidea, Papilionoidea) species with a wide latitudinal distribution in Europe. I started out with the list of Macrolepidoptera species that have been found in Finland (1040 species in total; as in Silvonen et al. 2014). For a species to be included, its distribution range in Europe had to exceed 60° latitude in the north (≈ southern border of Finland), and 45° latitude in the south.

Species not reaching 45° latitude in the south as well as those with clearly disjunct distribution ranges within Europe (which usually means that the southern populations are restricted to mountainous regions) were not considered (163 species in total). Additionally, I also excluded all migratory and occasionally occurring species (71 and 44 species, respectively). The final list of species (731 in total) included only species forming resident populations in Finland. The latitudinal range (15 latitudinal degrees, about 1650 km) considered corresponds to a 6.5–8°C difference in mean annual temperatures (Helsinki, Finland 60°10'N 24°56'E: +5.9°C; selected European cities close to 45°: +12.3°C to +13.8°C). This latitudinal difference in mean annual temperatures is roughly comparable to long-term projections of climate warming (IPCC 2018). The actual latitudinal distribution ranges (and thus temperatures and day lengths experienced by different populations) of most species substantially exceed this minimum range required for species inclusion. Given the wide range of season lengths over the latitudinal range considered, a species with a single generation per year throughout its whole European range was assumed to be obligately univoltine and retain a single generation per year with ongoing climate warming.

Species data

For each species considered, ecological data were retrieved from the most comprehensive (series of) monographs on European Macrolepidoptera available, with the series 'Noctuidae Europeae' (vol. 1–13, 1990–2011) and 'The Geometrid moths of Europe' (vol. 1–6, 2001–2019) providing data for the highest number of species (see 'Data sources', for a full list of data sources; also some unpublished and expert data were used). The voltinism (= number of generations per year) of a species, the primary response variable in the analyses, was ascertained separately for the whole European distribution range of each species, and for its populations inhabiting latitudes higher than 60° (mainly based on Finnish data from Silvonen et al. 2014). In the analyses, all species were classified either as univoltine species or multivoltine species. A species was considered univoltine if it has one generation per year throughout its European distribution range. A few species with a biennial (= semivoltine) life-cycle, i.e. species requiring more than one year for development, were also included in this category. By contrast, a species having two or more generations per year anywhere in its European distribution was considered multivoltine. Similarly, a species having a partial second generation anywhere in its range (i.e. some part of the population having a single generation whereas some part being bivoltine) was considered to have a potential for multivoltinism, and was thus classified as a multivoltine species. Note that there are very few species among the lepidopterans studied that have a multivoltine life cycle throughout their whole distribution range. At least the northernmost populations of the studied species are nearly always univoltine (or, in some cases, biennial).

As potential determinants of voltinism, I considered a set of traits that have previously been shown to be associated

with species' phenologies (Altermatt 2010b, Diamond et al. 2011, Végvári et al. 2015): overwintering stage, larval diet breadth, host type and body size. 1) Based on overwintering stage, species were grouped as egg, larval, pupal or adult overwinterers. Close to 30 species reported to overwinter as fully formed adults within their pupal case (known also as pupal imagoes) were classified as adult overwinterers. This was done so because, unlike other ('true') pupal overwinterers and similar to adult overwinterers, pupal imagoes do not require extra time for pupal development in spring. In the overwhelming majority of Lepidoptera, overwintering stage is fixed and species-specific. However, two species, *Pararge aegeria* and *Erebia ligea*, reported to overwinter in different stages, were left unclassified. 2) Five categories were distinguished on the basis of host type used in the larval stage. A species was classified as a forb-feeder, if it feeds exclusively or mostly on non-graminoid herbs (= forbs) or/and low-growing prostrate shrubs (such as *Vaccinium*, *Thymus* or *Calluna*). A few polyphagous species feeding both on forbs and grasses were also classified as forb-feeders. However, a species strictly specialized to feed on graminoid herbs (= Poaceae, Cyperaceae and Juncaceae) was classified as a grass-feeder. Species feeding exclusively or mostly on woody plants (excluding low-growing prostrate shrubs) were classified as tree-feeders. A separate category was established for species feeding both on trees and herbs. Lichen-feeders were assigned a separate category, but due to their small number they could not be included in the main analyses. 3) Three categories were distinguished to quantify larval diet breadth: monophagous (species feeding on plants of a single genus), oligophagous (species feeding on several host plant genera of a single family, or on a few genera of a few families) or polyphagous (species feeding on many host genera of different host families; see Holm et al. 2018, for an identical approach). 4) Average wingspan (maximum distance from one forewing tip to the other forewing tip), calculated as an average of mean male and female wingspans, was used as the measure of body size. For a few species with short-winged females, this measure was based on male data only.

Data analysis

Ecological correlates of the species' potential for multivoltinism were evaluated using generalized linear mixed models with logit as the link function. Voltinism was used as a binary response variable (univoltine versus multivoltine) in the analyses. Larval diet breadth (mono- versus oligo- versus polyphagous), host plant type (woody plants, forbs, grasses, woody + herbaceous plants) and overwintering stage (egg, larva, pupa, adult) entered the full model as categorical explanatory variables, and body size as a continuous variable. Due to missing combinations of categorical variables in the data set, interactions could not be included in the model, but were subjected to detailed scrutiny using descriptive statistics. Taxonomic hierarchy (superfamily/family/subfamily/genus; as in Silvonen et al. 2014), used as a surrogate to account for phylogenetic relationships (detailed

information on phylogenetic relationships is still missing for most Lepidoptera), was included as a nested random effect in all models.

Akaike's information criteria were used for model selection and multimodel inference. The models with $\Delta AIC_C \leq 2$ were considered to be close to the best model (Burnham and Anderson 2002). To infer the relative importance of predictor variables, model averaging was conducted across models with all possible combinations of variables. The relative importance of each variable was calculated by summing Akaike weights across all the models containing that particular variable. The larger the sum for a particular variable, the more important it was considered relative to other variables (Burnham and Anderson 2002).

All analyses were performed with R ver. 3.5.1 (<www.r-project.org>). The models were fit using the glmer function with the bobyqa algorithm for optimisation in 'lme4' package (Bates et al. 2015). The 'MuMIn' package was employed for model selection and averaging (Bartoni 2018). The r.squaredGLMM function in the same package was used to estimate variance explained (R^2) by fixed and random effects.

Results

General patterns

Almost half (47.1%; 344 species) of the total of 731 widely distributed moth and butterfly species examined have just one generation per year (univoltine life-cycle) throughout their European range (Supplementary material Appendix 1 Table A1). In northern populations (Finland, above 60°N), about two thirds of all species considered are univoltine, confirming the general latitudinal decrease in voltinism to the north (discussed in Zeuss et al. 2017).

The probability of being invariably univoltine across latitudes appears to be associated with species traits, with certain trait values strongly lowering the species' potential for multivoltinism. The model with the highest support (Akaike weight=0.61, altogether 16 models) contained species size (relative importance=1), overwintering stage (RI=1) and host type (RI=0.74) as predictors of voltinism. No other model received comparable support (i.e. $\Delta AIC_C \leq 2$). Altogether, the fixed traits in the best model (i.e. species size, overwintering stage, host type) explained 44% of the variation in voltinism. The cross-model relative importance of larval diet breadth was low (RI=0.21), and this variable did not appear in the best model.

The best model including the fixed effects and taxonomic hierarchy as a nested random effect was highly significant when compared to the model without taxonomic hierarchy, indicating a phylogenetic signal in the distribution of voltinism across the studied taxa (likelihood ratio test: $\chi^2=87.5$, $df=4$, $p<0.00001$). Taxonomic hierarchy, used to account for phylogenetic relationships in the model, explained 28% of the variation in voltinism. Nevertheless, uni- and multivoltine species show no strong clustering across the lepidopteran

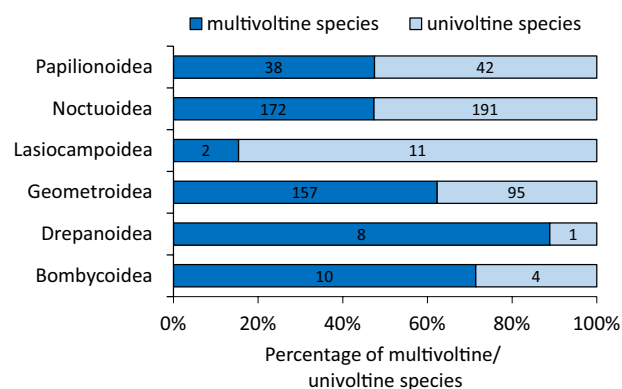


Figure 1. Proportions of multivoltine and univoltine species in different superfamilies of Macrolepidoptera. Numbers on the bars refer to the numbers of species in the respective taxa.

phylogeny: both univoltinism and multivoltinism occur in all superfamilies (Fig. 1) as well as in most families. Among the twelve families containing more than three species, Spingidae and Nolidae are the only exceptions in this regard: in these families, all studied species are capable for multiple generations per year. Variation in voltinism among species is also common within individual subfamilies and genera (Supplementary material Appendix 1 Table A1).

Species size

Species size has a significant effect on its potential for multivoltinism (Table 1). Across species, the average wingspan of multivoltine species is 12.8% smaller than in univoltine species (mean $\pm$ SE: multivoltine species 31.3 ± 0.6 mm; univoltine species 35.9 ± 0.6 mm). This pattern is qualitatively consistent across major superfamilies (Papilionoidea, Geometroidea, Noctuoidea; Fig. 2) as well as at all levels of all categorical variables (i.e. overwintering stage, host type, larval diet breadth; Fig. 3), with the exception of pupal overwinterers. This exception can mostly be attributed to a high number of *Eupithecia* species among univoltine pupal overwinterers, a species-rich genus of very small moths. Without *Eupithecia*, univoltine species have a 16.6% larger wingspan also among pupal overwinterers.

Table 1. Results of the best generalized linear mixed model explaining the effects of species traits on the species' potential for multivoltinism. Taxonomic hierarchy (superfamily/family/subfamily/genus) was included as a nested random effect to account for phylogenetic non-independence of the data. Overall, the selected model was highly significant when compared to the null model with only the random effect included (likelihood ratio test: $\chi^2=174.9$, $df=7$, $p<0.00001$).

Variable	χ^2	df	p
Overwintering stage	60.8	3	<0.00001
Host type	7.7	3	0.051
Species size	18.6	1	0.00002

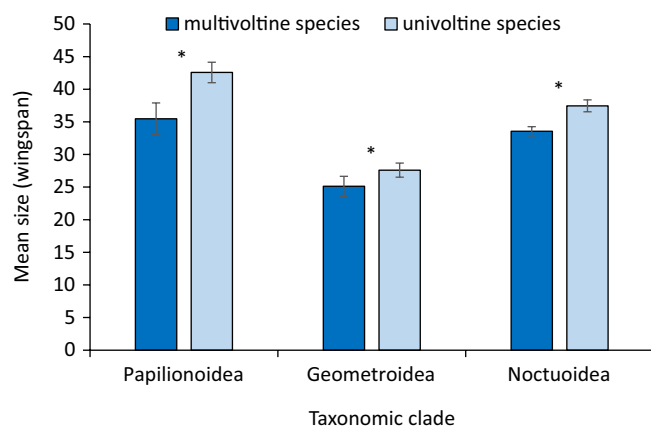


Figure 2. Mean size (= wingspan) of multivoltine and univoltine species in major superfamilies of the studied Macrolepidoptera. Error bars denote standard errors. Asterisks indicate significant differences (adjusted for multiple testing).

Overwintering stage

Among the ecological traits examined, overwintering stage has the strongest effect on the species' potential for multivoltinism (Table 1). The proportion of multivoltine species gradually increases from species overwintering as eggs to species overwintering as adults, larvae and pupae (Fig. 4). The respective patterns within three major superfamilies largely coincide with the general pattern (Supplementary material Appendix 1 Fig. A1). In species overwintering as eggs, multivoltinism is remarkably rare: just 8.8% of such species are fully or partly multivoltine at least in some part of their European distribution range. The proportion of multivoltine species among egg overwinterers is low irrespective of host type and larval diet breadth (Supplementary material Appendix 1 Table A2, A3).

The proportion of multivoltine species is also low in species overwintering as adults: about a quarter of these species (25.5%) have a partly or fully multivoltine life cycle anywhere in Europe (Fig. 4). The proportion of multivoltine species in the subset of adult overwinterers passing the winter as fully formed adults within their pupal case (pupal imagoes) was similar to the proportion of multivoltine species among the rest of adult overwinterers (28.6% and 22.2%, respectively). Univoltine species among adult overwinterers do not exceed one-third in any host type and larval diet breadth category, except for in the species-poor category of monophagous adult overwinterers (Supplementary material Appendix 1 Table A2, A3).

By contrast, multivoltine species comprise an overwhelming majority of pupal overwinterers: more than four fifths (81.0%) are partly or fully multivoltine within their European distribution range (Fig. 4). The proportion of multivoltine species among pupal overwinterers is uniformly high irrespective of host type and larval diet breadth (always $\geq 75\%$; Supplementary material Appendix 1 Table A2, A3). Species overwintering as larvae fall between these extremes, with

approximately half of such species (50.7%) being multivoltine (Fig. 4; Supplementary material Appendix 1 Table A2, A3).

Host type

There is also a tendency, although marginally non-significant, for the species' potential for multivoltinism to vary with host type (Table 1). The gap in the proportions of multivoltine species is widest between species feeding on graminoid plants (grasses, sedges and rushes), and those feeding on forbs and low-growing prostrate shrubs (Fig. 5). In the latter, 62.1% of the species are multivoltine, whereas in species feeding on graminoid plants just 28.2% of the species are multivoltine. Twenty-one species feeding internally within grass stems are univoltine with just one exception. Species feeding on woody plants or both on woody and herbaceous plants fall between these two figures, with multivoltinism occurring in 52.9% and 41.5% species, respectively (Fig. 5). Species feeding exclusively on coniferous trees have a somewhat higher incidence of multivoltinism (61.1% of 18 species) than other species feeding on woody plants. In lichen-feeding species, 40% of the 20 species have a multivoltine life cycle.

Larval diet breadth

Larval diet breadth shows no overall association with voltinism (Table 1). In all three categories, i.e. mono-, oligo- and polyphages, multivoltine species comprise slightly more than half of all species (56.2, 54.2 and 51.2%, respectively).

Discussion

The increase in the number of generations per year (i.e. voltinism) is one of the most commonly reported phenological responses to climate warming in insects (Altermatt 2010a, Pöyry et al. 2011). Nevertheless, the results of this study suggest that a considerable proportion of insect species are likely to maintain univoltine phenology despite increasing temperatures. The examination of 731 European moth and butterfly species with a wide latitudinal distribution (15° as a minimum) showed that only slightly more than half (52.9%) of these species are multivoltine in at least some part of their range. The rest of species have a single generation throughout their distribution range despite a high latitudinal variation in temperature and season length. My results show that a substantial proportion of the among-species variation in voltinism can be explained by species traits.

Focusing on the subset of (latitudinally) widely distributed species allowed me to evaluate their potential for multivoltinism in response to the levels of climate warming far beyond what we have experienced to date ($+1^\circ\text{C}$ increase; IPCC 2018), and comparable to long-term climate projections (IPCC 2018). It is very likely that most of the species that are univoltine across a latitudinal gradient of more than 1600 km (corresponding to at least 6.5°C difference

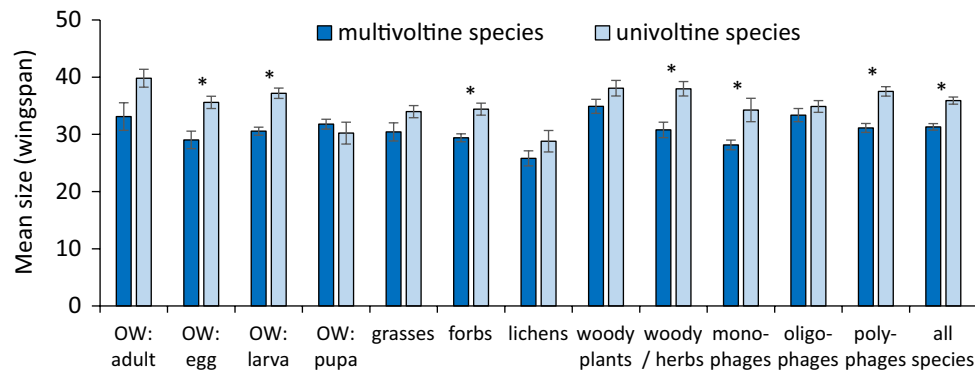


Figure 3. Mean size (= wingspan) of multivoltine and univoltine species among the studied Macrolepidoptera classified by overwintering stage (OW), host type and larval diet breadth (see Material and methods for details of classification). Error bars denote standard errors. Asterisks indicate significant differences (adjusted for multiple testing).

in mean annual temperatures and large variation in season length) represent cases of 'true' (= obligatory) univoltinism. Such species are unlikely to become multivoltine also with further increase in temperatures and season length. Observed changes in voltinism in response to more limited warming experienced to date are certainly less likely to reveal true cases of univoltinism.

In insects capable for multiple generations in a season, voltinism typically decreases with latitude (Zeuss et al. 2017; this study). In this light, having a single generation per year across a wide latitudinal temperature gradient indicates the presence of some constraint that makes additional generations either impossible or at least strongly disadvantageous. The scattered distribution of univoltine and multivoltine species across the lepidopteran phylogeny suggests that phylogenetic constraints do not strongly limit changes in voltinism,

and that the trait is open to ecologically-driven adaptive evolution. Indeed, the occurrence of univoltine and multivoltine species is common even within individual genera.

Quite obviously, all else being equal, one might expect larger species to be more time-constrained to produce multiple generations per year. The collected data show that body size indeed has a negative effect on the species' potential for multivoltinism. Univoltine species are larger, on average, than multivoltine species in nearly all ecological categories, irrespective of whether species are classified according to overwintering stage, host type or larval diet breadth. Similarly, univoltine species are larger in all major taxonomic clades. The results regarding the general body size–voltinism relationship are thus consistent with the findings of earlier studies (Horne et al. 2015, Zeuss et al. 2017). Nevertheless, my results show that body size is far from being an absolute constraint to having multiple generations per year. In my sample of species, just four of the ten largest species are univoltine

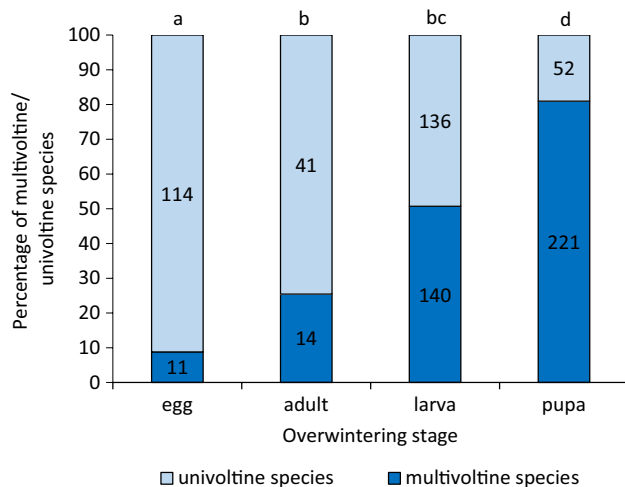


Figure 4. Proportions of multivoltine and univoltine species among the studied Macrolepidoptera classified by overwintering stage. Different letters above the bars indicate statistically significant differences in the proportions of uni- and multivoltine species between respective categories ($p < 0.05$; Tukey post hoc test for multiple comparisons). Numbers on the bars refer to the numbers of species in the respective categories.

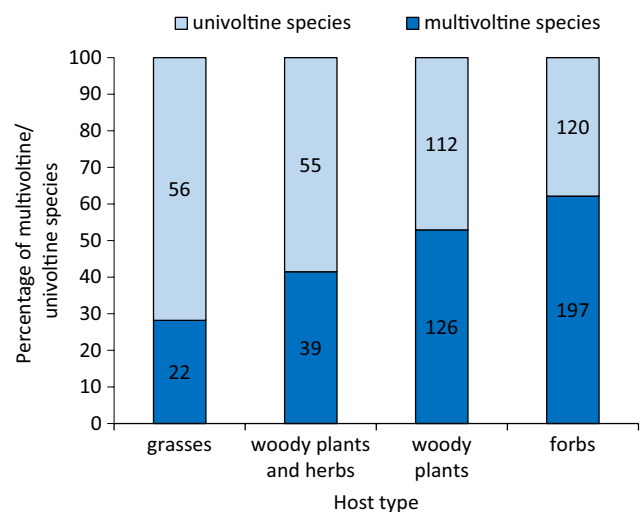


Figure 5. Proportions of multivoltine and univoltine species among the studied Macrolepidoptera feeding on different host types (see Material and methods for details of classification). Numbers on the bars refer to the numbers of species in the respective categories.

throughout their European range, with others being multivoltine at least in southern parts of their distribution. At the opposite end of the size gradient, only six of the ten smallest species are multivoltine. Species size alone thus excludes neither uni- nor multivoltinism, and other species' characteristics can be expected to play a role.

The analysis indicates that the species' potential for additional generations per year is strongly related to the factors that determine the overwintering stage of the species. The contrast is particularly strong between pupal overwinterers (81% multivoltine), on the one hand, and egg (9% multivoltine) and adult (26% multivoltine) overwinterers, on the other hand. The life-histories of a considerable share of species in the two latter groups are typical of so-called spring- (or flush-) feeders (Hunter and Elkinton 2000, Van Asch and Visser 2007). Their eggs hatch when buds of woody plants begin to burst in the spring (adult overwinterers lay their eggs before bud burst), and the larvae are adapted to feed on bursting buds and young foliage of trees and shrubs, taking advantage of their high nitrogen content and low levels of defences. Because of declining nutritional quality and increasing defensive chemicals, maturing leaves become increasingly inedible for these species, with detrimental effects on larval growth and survival (Parry et al. 1998, Hunter and Elkinton 2000, Tikkanen and Lyytikäinen-Saarenmaa 2002). Additional generations are therefore strongly disfavoured in spring-feeders on trees and shrubs.

Species feeding strictly on grasses form another major ecological group of lepidopterans with a low incidence of multivoltinism. Similar to most egg-overwintering spring-feeders on trees, all grass-feeders overwintering as eggs are univoltine. Nevertheless, multivoltinism is also infrequent among grass-feeders overwintering as larvae (28% multivoltine). The low incidence of multivoltine species among grass-feeders is plausibly related to the nutritional quality and seasonal dynamics of graminoid plants. Grasses typically contain primary nutrients at much lower concentrations than forbs and woody plants (Bernays and Barbehenn 1987, Tscharrntke and Greiler 1995). Moreover, while in general phenologically young plants are nutritionally superior for herbivorous insects (Scriber and Slansky 1981), grasses are known to deteriorate in nutritional quality particularly rapidly (Bernays and Barbehenn 1987). The window of opportunity for strict grass-feeders may therefore be too short for additional generations to successfully develop, irrespective of the length of the season with favourable temperatures. The importance of host plant nutritional quality on voltinism has recently been demonstrated also by Pöyry et al. (2017), who showed that the number of generations per season increases with foliar nitrogen content of host plants. Endophagous grass-feeders seem to be particularly restricted in voltinism: none of the 21 species feeding internally within grass stems has more than one generation per year. Besides nutritional constraints, their window of opportunity is further restricted by the availability of sufficiently thick stems in which they can feed. On the other hand, the few grass-feeders overwintering as pupa (five

species) seem to have overcome host plant related constraints: all these species have multiple generations per year.

Larval diet breadth has no discernible overall effect on the species' potential for multivoltinism. The proportion of multivoltine species among mono-, oligo- and polyphagous species is strikingly similar (51–56%). No doubt, in some cases, feeding on a single host plant species (or particular organs, such as fruits and flowers) that is available for a short period, eliminates the opportunity for multiple generations (e.g. *Parnassius mnemosyne* on *Corydalis* spp.: Välimäki and Itämes 2005; *Eupithecia immundata* on fruits of *Actaea spicata*: Ehrlén et al. 2015; *Anthocharis cardamines* on flowers and growing siliques of Brassicaceae: Posledovich et al. 2015). However, the majority of strictly-specialized species utilize host species with less-restricted seasonal availability. Another plausible but to my knowledge untested explanation for multivoltinism in monophagous species is that being adapted to feed on a single host plant has allowed them to speed up their development, and pass multiple generations despite a shorter period of host availability and/or seasonally decreasing nutritional quality of the host plant. It should also be noted that classifying species by larval diet breadth across a large geographic area is not as straightforward as for local scales. On the one hand, several species that could have been classified as mono- or oligophagous at the local scale (e.g. for Finnish populations), were actually classified as oligo- or polyphagous species because of their much broader diet at the European level. On the other hand, a long list of potentially acceptable hosts in polyphagous species does not preclude a considerable preference for particular host species (Pöldmaa et al. 2016).

The cross-species variation in the potential for increased voltinism with climate warming is likely to have considerable implications for population and community dynamics of Lepidoptera as well as associated lower and upper trophic levels (Altermatt 2010b). On the one side of the voltinism spectrum, the capacity for an increase in voltinism has been proposed to promote population growth, and increase the likelihood of outbreaks (Tobin et al. 2014; but see Hunter 1991). Recent studies have provided direct evidence that multivoltine lepidopterans in temperate and boreal environments are indeed generally more abundant (Mangels et al. 2017, Potocky et al. 2018), and show more positive population trends than do univoltine species (Pöyry et al. 2017, Macgregor et al. 2019; see also WallisDeVries and van Swaay 2017). On the opposite side of the spectrum, many univoltine species, particularly spring-feeders whose survival and development is strongly synchronized with their host phenology, are potentially vulnerable to phenological mismatch between interacting species that have been proposed to accompany climate warming (Visser and Holleman 2001, Forkner et al. 2008, Abarca and Lill 2015). Nevertheless, phenological advances have been most pronounced in egg- (Altermatt 2010b) and adult-overwintering Lepidoptera (Diamond et al. 2011), ecological categories in which the proportion of univoltine species is particularly high. This may indicate that they are relatively flexible in adjusting their phenology in response to variable

climatic conditions. Warmer conditions may pose a potential threat also for species capable to increase the number of generations. An increase in voltinism can be maladaptive if rapid environmental changes trigger developmental decisions (such as adding an extra generation) that convey high mortality – a phenomenon termed developmental trap (Pöyry et al. 2011, Van Dyck et al. 2015, Blackshaw and Esbjerg 2018). Given these multifaceted consequences and implications, accounting for differential voltinism responses may be one of the keys to address future challenges in biodiversity conservation and pest management.

Data availability statement

Data are available from the Dryad Digital Repository: <<http://dx.doi.org/10.5061/dryad.5hqbzkh31>> (Teder 2020).

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References

- Abarca, M. and Lill, J. T. 2015. Warming affects hatching time and early season survival of eastern tent caterpillars. – *Oecologia* 179: 901–912.
- Altermatt, F. 2010a. Climatic warming increases voltinism in European butterflies and moths. – *Proc. R. Soc. B* 277: 1281–1287.
- Altermatt, F. 2010b. Tell me what you eat and I'll tell you when you fly: diet can predict phenological changes in response to climate change. – *Ecol. Lett.* 13: 1475–1484.
- Bartoń, K. 2018. MuMIn: multi-model inference. – R package ver. 1.42.1. <<https://cran.r-project.org/web/packages/MuMIn/>>.
- Bates, D. et al. 2015. Fitting linear mixed-effects models using lme4. – *J. Stat. Softw.* 67: 1–48.
- Bernays, E. A. and Barbehenn, R. 1987. Nutritional ecology of grass foliage-chewing insects. – In: Slansky, F. and Rodriguez, J. G. (eds), *Nutritional ecology of insects, mites, spiders and related invertebrates*. Wiley, pp. 147–175.
- Blackshaw, R. P. and Esbjerg, P. 2018. Statistical indicators for insects caught in the development trap. – *J. Appl. Entomol.* 142: 272–276.
- Blois, J. L. et al. 2013. Space can substitute for time in predicting climate-change effects on biodiversity. – *Proc. Natl Acad. Sci. USA* 110: 9374–9379.
- Burnham, K. P. and Anderson, D. R. 2002. Model selection and multimodel inference: a practical information-theoretic approach. – Springer.
- Carter, S. K. et al. 2018. Shifts in phenological distributions reshape interaction potential in natural communities. – *Ecol. Lett.* 21: 1143–1151.
- Cohen, J. M. et al. 2018. A global synthesis of animal phenological responses to climate change. – *Nat. Clim. Change* 8: 224–228.
- Danilevsky, A. S. et al. 1970. Biological rhythms in terrestrial arthropods. – *Annu. Rev. Entomol.* 15: 201–244.
- Diamond, S. E. et al. 2011. Species' traits predict phenological responses to climate change in butterflies. – *Ecology* 92: 1005–1012.
- Duchenne, F. et al. 2020. Phenological shifts alter the seasonal structure of pollinator assemblages in Europe. – *Nat. Ecol. Evol.* 4: 115–121.
- Ehrlén, J. et al. 2015. Flowering schedule in a perennial plant; life-history tradeoffs, seed predation and total offspring fitness. – *Ecology* 96: 2280–2288.
- Forkner, R. E. et al. 2008. Timing is everything? Phenological synchrony and population variability in leaf-chewing herbivores of *Quercus*. – *Ecol. Entomol.* 33: 276–285.
- Forrest, J. and Miller-Rushing, A. J. 2010. Toward a synthetic understanding of the role of phenology in ecology and evolution. – *Phil. Trans. R. Soc. B* 365: 3101–3112.
- Fric, Z. F. et al. 2020. Phenology responses of temperate butterflies to latitude depend on ecological traits. – *Ecol. Lett.* 23: 172–180.
- Gardner, J. L. et al. 2011. Declining body size: a third universal response to warming? – *Trends Ecol. Evol.* 26: 285–291.
- Hodgson, J. A. et al. 2011. Predicting insect phenology across space and time. – *Global Change Biol.* 17: 1289–1300.
- Holm, S. et al. 2018. Reproductive behaviour indicates specificity in resource use: phylogenetic examples from temperate and tropical insects. – *Oikos* 127: 1113–1124.
- Horne, C. R. et al. 2015. Temperature–size responses match latitudinal–size clines in arthropods, revealing critical differences between aquatic and terrestrial species. – *Ecol. Lett.* 18: 327–335.
- Hunter, A. F. 1991. Traits that distinguish outbreaking and non-outbreaking Macrolepidoptera feeding on northern hardwood trees. – *Oikos* 60: 275–282.
- Hunter, A. F. and Elkinton, J. S. 2000. Effects of synchrony with host plant on populations of a spring-feeding Lepidopteran. – *Ecology* 81: 1248–1261.
- IPCC 2018. Global warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development and efforts to eradicate poverty (Masson-Delmotte, V. et al. eds). – IPCC.
- Kharouba, H. M. et al. 2014. Predicting the sensitivity of butterfly phenology to temperature over the past century. – *Global Change Biol.* 20: 504–514.
- Kharouba, H. M. et al. 2018. Global shifts in the phenological synchrony of species interactions over recent decades. – *Proc. Natl Acad. Sci. USA* 115: 5211–5216.
- Macgregor, C. J. et al. 2019. Climate-induced phenology shifts linked to range expansions in species with multiple reproductive cycles per year. – *Nat. Comm.* 10: 4455.
- Mangels, J. et al. 2017. Diversity and trait composition of moths respond to land-use intensification in grasslands: generalists replace specialists. – *Biodivers. Conserv.* 26: 3385–3405.
- Parmesan, C. 2006. Ecological and evolutionary responses to recent climate change. – *Annu. Rev. Ecol. Evol. Syst.* 37: 637–669.
- Parmesan, C. 2007. Influences of species, latitudes and methodologies on estimates of phenological response to global warming. – *Global Change Biol.* 13: 1860–1872.
- Parmesan, C. et al. 1999. Poleward shifts in geographical ranges of butterfly species associated with regional warming. – *Nature* 399: 579–583.

- Parry, D. et al. 1998. Budbreak phenology and natural enemies mediate survival of first-instar forest tent caterpillar (Lepidoptera: Lasiocampidae). – *Environ. Entomol.* 27: 1368–1374.
- Pöldmaa, K. et al. 2016. Polyphagy on unpredictable resources does not exclude host specialization: insects feeding on mushrooms. – *Ecology* 97: 2824–2833.
- Posledovich, D. et al. 2015. Latitudinal variation in diapause duration and post-winter development in two pierid butterflies in relation to phenological specialization. – *Oecologia* 177: 181–190.
- Potocky, P. et al. 2018. Life-history traits of central European moths: gradients of variation and their association with rarity and threats. – *Insect Conserv. Divers.* 11: 493–505.
- Pöyry, J. et al. 2011. Climate-induced increase of moth multivoltinism in boreal regions. – *Global Ecol. Biogeogr.* 20: 289–298.
- Pöyry, J. et al. 2017. The effects of soil eutrophication propagate to higher trophic levels. – *Global Ecol. Biogeogr.* 26: 18–30.
- Primack, R. B. et al. 2009. Spatial and interspecific variability in phenological responses to warming temperatures. – *Biol. Conserv.* 142: 2569–2577.
- Roff, D. 1980. Optimizing development time in a seasonal environment: the ‘ups and downs’ of clinal variation. – *Oecologia* 45: 202–208.
- Roy, D. B. et al. 2015. Similarities in butterfly emergence dates among populations suggest local adaptation to climate. – *Global Change Biol.* 21: 3313–3322.
- Scriber, J. M. and Slansky, F. 1981. The nutritional ecology of immature insects. – *Annu. Rev. Entomol.* 26: 183–211.
- Silvonen, K. et al. 2014. A field guide to the butterflies and moths of Finland. – Bugbook Publishing.
- Tammaru, T. et al. 2001. Autumnal moth – why autumnal? – *Ecol. Entomol.* 26: 646–654.
- Tauber, M. J. et al. 1986. Seasonal adaptations of insects. – Oxford Univ. Press.
- Teder, T. 2020. Data from: Phenological responses to climate warming in temperate moths and butterflies: species traits predict future changes in voltinism. – Dryad Digital Repository, <<http://dx.doi.org/10.5061/dryad.5hqbzkh31>>.
- Tikkanen, O. P. and Lyytikäinen-Saarenmaa, P. 2002. Adaptation of a generalist moth, *Operophtera brumata*, to variable budburst phenology of host plants. – *Entomol. Exp. Appl.* 103: 123–133.
- Tobin, P. C. et al. 2014. The influence of climate change on insect invasions in temperate forest ecosystems. – In: Fenning, T. (ed.), *Challenges and opportunities for the World's forests in the 21st century*. Forestry sciences, vol 81. Springer, pp. 267–293.
- Tscharntke, T. and Greiler, H.-J. 1995. Insect communities, grasses and grasslands. – *Annu. Rev. Entomol.* 40: 535–558.
- Välimäki, P. and Itämes, J. 2005. Effects of canopy coverage on the immature stages of the clouded apollo butterfly [*Parnassius mnemosyne* (L.)] with observations on larval behaviour. – *Entomol. Fennica* 16: 117–123.
- Valtonen, A. et al. 2011. Environmental controls on the phenology of moths: predicting plasticity and constraint under climate change. – *Oecologia* 165: 237–248.
- Van Asch, M. and Visser, M. E. 2007. Phenology of forest caterpillars and their host trees: the importance of synchrony. – *Annu. Rev. Entomol.* 52: 37–55.
- Van Dyck, H. et al. 2015. The lost generation hypothesis: could climate change drive ectotherms into a developmental trap? – *Oikos* 124: 54–61.
- Végvári, Z. et al. 2015. Life-history traits and climatic responsiveness in noctuid moths. – *Oikos* 124: 235–242.
- Visser, M. E. and Holleman, L. J. M. 2001. Warmer springs disrupt the synchrony of oak and winter moth phenology. – *Proc. R. Soc. B* 268: 289–294.
- Visser, M. E. and Gienapp, P. 2019. Evolutionary and demographic consequences of phenological mismatches. – *Nat. Ecol. Evol.* 3: 879–885.
- WallisDeVries, M. F. and van Swaay, C. A. M. 2017. A nitrogen index to track changes in butterfly species assemblages under nitrogen deposition. – *Biol. Conserv.* 212: 448–453.
- Zeuss, D. et al. 2017. Environmental drivers of voltinism and body size in insect assemblages across Europe. – *Global Ecol. Biogeogr.* 26: 154–165.

Supplementary material (available online as Appendix oik-07119 at <www.oikosjournal.org/appendix/oik-07119>). Appendix 1.

Data sources

- Ahola, M. and Silvonen, K. 2005. Larvae of northern European Noctuidae, Volume 1. – Apollo Books.
- Ahola, M. and Silvonen, K. 2008. Larvae of northern European Noctuidae, Volume 2. – Apollo Books.
- Ahola, M. and Silvonen, K. 2010. Larvae of northern European Noctuidae, Volume 3. – Apollo Books.
- Bartsch, D. 1997. Die Schmetterlinge Baden-Württembergs Band 5: Nachtfalter III. – Eugen Ulmer Verlag.
- Ebert, G. 2001. Die Schmetterlinge Baden-Württembergs Band 8: Nachtfalter VI. – Eugen Ulmer Verlag.
- Ebert, G. 2003. Die Schmetterlinge Baden-Württembergs Band 9: Nachtfalter VII. – Eugen Ulmer Verlag.
- Ebert, G. 2005. Die Schmetterlinge Baden-Württembergs Band 10: Ergänzungsband. – Eugen Ulmer Verlag.
- Ebert, G. and Rennwald, E. 1991. Die Schmetterlinge Baden-Württembergs Band 1: Tagfalter I Papilionidae, Pieridae, Nymphalidae. – Eugen Ulmer Verlag.
- Ebert, G. and Rennwald, E. 1991. Die Schmetterlinge Baden-Württembergs Band 2: Tagfalter II Satyridae, Libytheidae, Lycaenidae, Hesperidae. – Eugen Ulmer Verlag.
- Ebert, G. and Rennwald, E. 1994. Die Schmetterlinge Baden-Württembergs Band 3: Nachtfalter I Spinnerartige. – Eugen Ulmer Verlag.
- Ebert, G. and Rennwald, E. 1994. Die Schmetterlinge Baden-Württembergs Band 4: Nachtfalter II Eulenartige. – Eugen Ulmer Verlag.
- Fibiger, M. 1990. Noctuidae Europaeae, vol. 1: Noctuinae I. – Apollo Books.
- Fibiger, M. 1993. Noctuidae Europaeae, vol. 2: Noctuinae II. – Apollo Books.
- Fibiger, M. 1997. Noctuidae Europaeae, vol. 3: Noctuinae III. – Apollo Books.
- Fibiger, M. and Hacker, H. 2007. Noctuidae Europaeae, vol. 9: Amphipyridae, Condiciidae, Eriopinae, Xyleninae. – Apollo Books.
- Fibiger, M. et al. 2009. Noctuidae Europaeae, vol. 11: Nolinae (s.l.), Heliethinae, Metoponinae, Eustrothiinae, Bagisarininae, Pantheinae, Raphaninae, Dilobinae, Acronyctinae, Bryophilinae. – Apollo Books.
- Fibiger, M. et al. 2010. Noctuidae Europaeae, vol. 12: Rivulinae, Boletobiinae, Hypenodinae, Aracopterinae, Eublemminae, Herminiinae, Hypeninae, Phytometrinae, Euteliinae and Micronoctuidae. Including Supplement to volumes 1–11. – Apollo Books.
- Goater, B. et al. 2003. Noctuidae Europaeae, vol. 10: Catocalinae & Plusiinae. – Apollo Books.
- Hacker, H. et al. 2002. Noctuidae Europaeae, vol. 4: Hadeninae I. – Apollo Books.
- Hausmann, A. 2001. The geometrid moths of Europe, Vol. 1 Introduction, Archiarinae, Orthostixinae, Desmobathrinae, Alsophilinae, Geometrinae. – Apollo Books.
- Hausmann, A. and Wilson, D. 2004. The geometrid moths of Europe, Vol. 2 Sterrhinae. – Apollo Books.
- Hausmann, A. et al. 2012. The geometrid moths of Europe, Vol. 3 Larentiinae I. – Apollo Books.
- Mironov, V. et al. 2003. The geometrid moths of Europe, Vol. 4 Larentiinae II (Perizomini and Eupitheciini). – Apollo Books.
- Müller, B. et al. 2019. The geometrid moths of Europe, Vol. 6 Part 1–2 Ennominae II (Boarmiini, Gnophini, additions to previous volumes). – Brill.
- Ronkay, L. et al. 2001. Noctuidae Europaeae, vol. 5: Hadeninae II. – Apollo Books.
- Ronkay, L. and Ronkay, G. 1994. Noctuidae Europaeae, vol. 6: Cucullinae I. – Apollo Books.
- Ronkay, G. and Ronkay, L. 1995. Noctuidae Europaeae, vol. 7: Cucullinae II. – Apollo Books.
- Silvonen, K. et al. 2014. A field guide to the butterflies and moths of Finland. – Bugbook Publishing.
- Skou, P. et al. 2015. The geometrid moths of Europe, Vol. 5 Ennominae I. – Apollo Books.
- Steiner, A. and Ebert, G. 1998. Die Schmetterlinge Baden-Württembergs Band 6: Nachtfalter IV Spezieller Teil: Noctuidae (Forsetzung). – Eugen Ulmer Verlag.
- Steiner, A. and Ebert, G. 1998. Die Schmetterlinge Baden-Württembergs Band 7: Nachtfalter V Spezieller Teil: Noctuidae (Forsetzung). – Eugen Ulmer Verlag.
- Tolman, T. and Lewington, R. 2009. Collins butterfly guide: the most complete guide to the butterflies of Britain and Europe. – HarperCollins.
- Witt, T. J. and Ronkay, L. 2011. Noctuidae Europaeae, vol. 13: Lymantriinae and Arctiinae. Including Phylogeny and checklist of the Quadrid Noctuoidea of Europe. – Apollo Books.
- Zilli, A. et al. 2005. Noctuidae Europaeae, vol. 8: Apameini. – Apollo Books.